ICT2122 · Object Oriented Programming

Foundations: Objects, State, and Behavior

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

An object combines state with operations that use or update that state. A class defines a family of objects. Encapsulation keeps representation details private and exposes a small interface that preserves valid state.

2. Example and reasoning

A bank-account object can keep its balance private and expose deposit and withdrawal operations. The methods validate amounts so callers cannot set an impossible balance directly.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “foundations: objects, state, and behavior” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.